

Thank you for purchasing our Ariel Voice Generation asset.

This document will show you how to use the asset in Unreal Engine.

If you have any question, contact us at : contact@xandimmersion.com

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1. Prerequisites

First, please download the latest version of Ariel plugin:

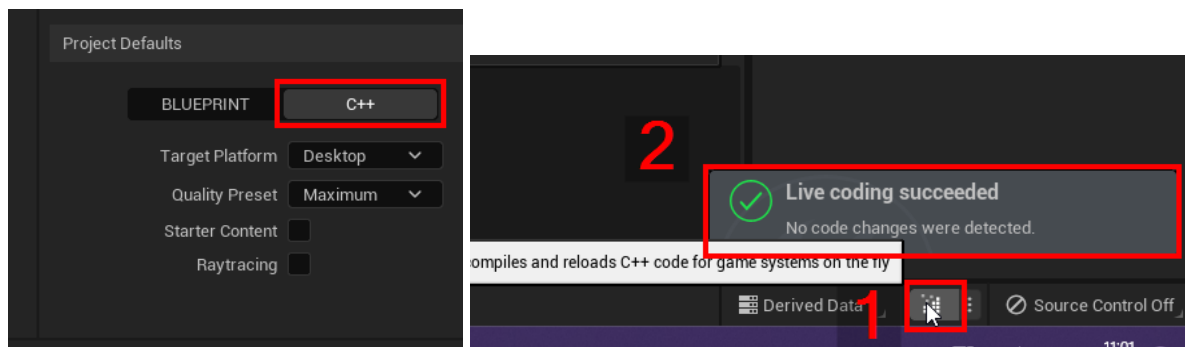
<https://drive.google.com/drive/folders/1ryhkEGTy0vQdmWG40f3p1hOkO-SbX17u>

There is three different version of the project:

- Plugin + demo project: Contains the full Unreal project, already compiled.
- Plugin only: Contains only the Ariel Unreal plugin, compiled.
- Plugin not compiled *: Contains only the Ariel Unreal plugin, not compiled.

*** You must be able to compile C++ code for Unreal Engine. For that, you need to have the Visual Studio build tools for Unreal Engine installed as well as the .NET SDK ! Find more information on how to set up Visual studio for Unreal engine [here](#).**

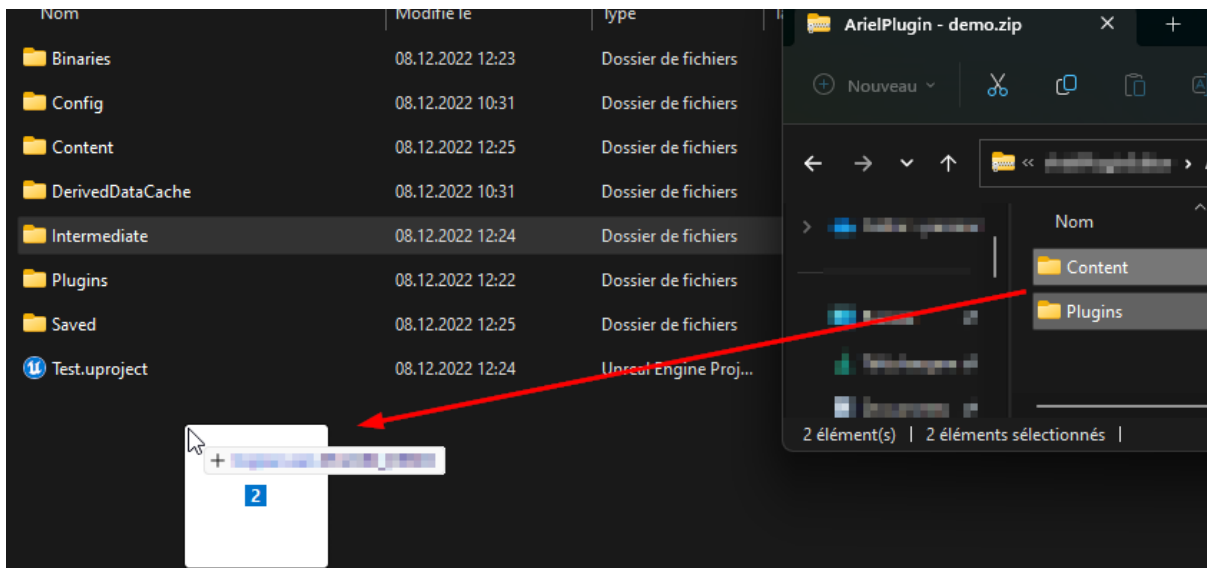
To ensure that you have all build tools installed, try to create a new C++ Unreal Project and make sure that the Editor starts correctly. Once the editor is open, click on the compile button on the bottom right corner and ensure that the compilation succeeded:



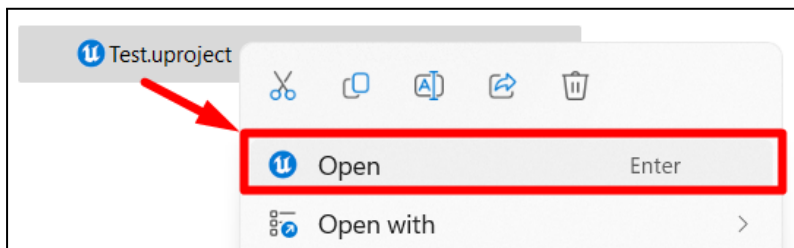
2. How to install the Ariel plug-in

First, close the unreal engine project where you want Ariel to be installed !

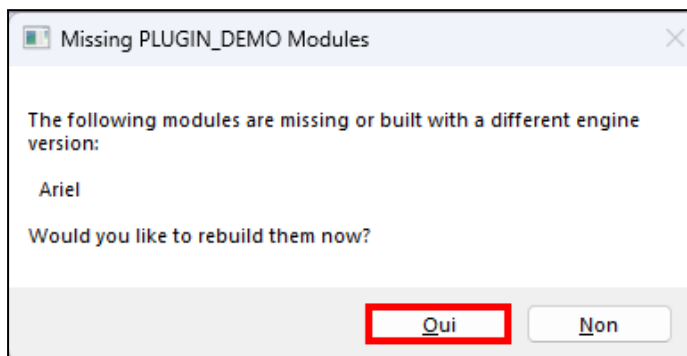
Extract the content from the zip archive and paste it to your base project directory:



Then, reopen your Unreal project (Right click on <Project_name>.uproject ⇒ Open):



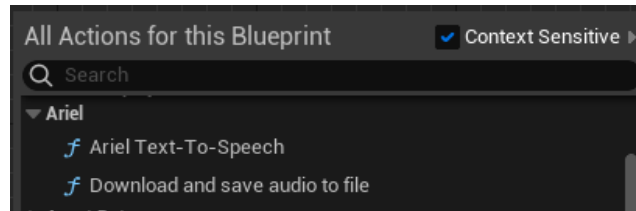
Optional: If the “Missing Modules” Window appears, choose “Yes”:



IMPORTANT: You will need to have the Visual Studio build tools for Unreal Engine installed in order to rebuild missing modules! See prerequisites for more information.

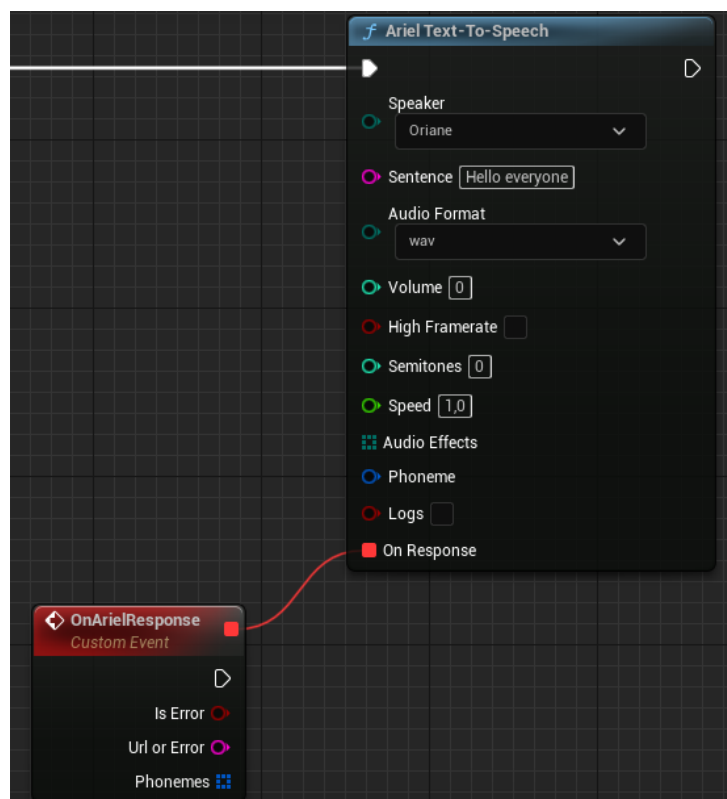
3. How to use the Ariel plug-in in Blueprints (Runtime & Editor)

The plugin have 2 functions that can be used directly in Blueprints:



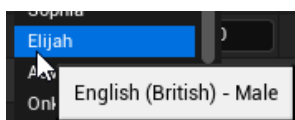
3.1. Ariel Text-To-Speech

This function calls the Ariel API using HTTPS request with all defined Ariel parameters (speaker, sentence, format, effects, ...). The response is sent through a delegate :



More information about Ariel function parameters:

Speaker: Defines the voice used to play the sentence. Hover the cursor on an item on the list to see more details about it. **Only the voice “Zenaya” is available in the free version!**



⇒ You can find the list of all speakers at the end of this document!

Sentence: Defines the sentence that the API needs to generate the speech for.

Audio format: Defines the format of the audio played. Supported formats are wav, mp3, mp4 and flac.

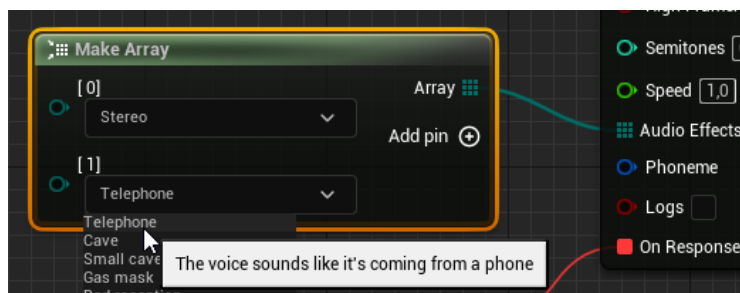
Volume: Adjust the volume of the speaker by the specified dB, i.e: +3dB, -5dB.

High framerate: Generate a speech using 44.1 KHz frequency instead of 22.05 KHz (default).

Semitones: Defines the pitch of the generated speech as semitones (so 1 octave is +-12).

Speed: Defines the speed generated speech. Normal speed is 1.0.

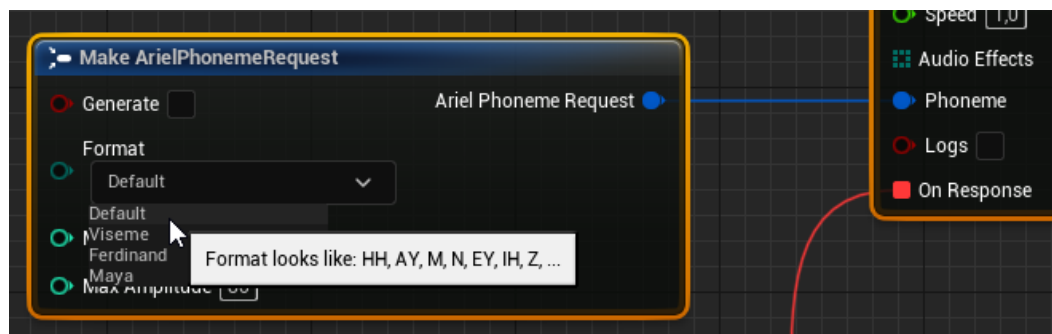
Audio effects: Defines all effects who will be applied to the audio. You can combine effects using an array with multiple values. Hover the cursor on an item on the list to see more details about it.



Available effects:

- Telephone
- Cave
- Small cave
- Gas mask
- Bad reception
- Next room
- Stereo

Phonemes: Indicate how the phonemes need to be generated.



Generate: Indicate if the phonemes should be generated or not.

Format: Defines the phoneme format. Hover the cursor for more details.

Min ampl: Defines the phonemes minimum amplitude (in dB).

Max ampl: Defines the phonemes maximum amplitude (in dB).

Logs: Indicate if the logs should be printed to the console (Dev only).

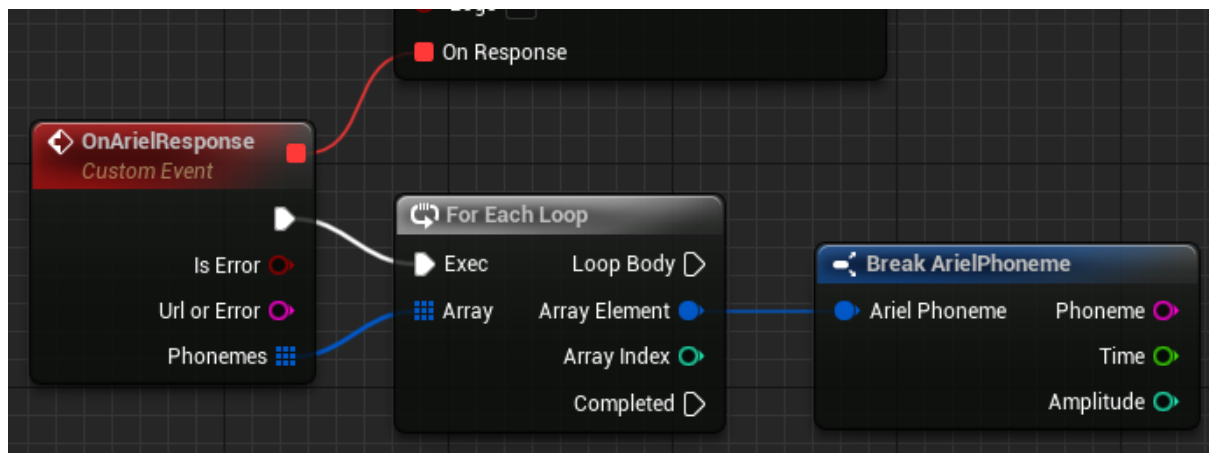
More information about Ariel return response:

You can bind and handle Ariel response using the node “Add custom event” or “Create Event”.

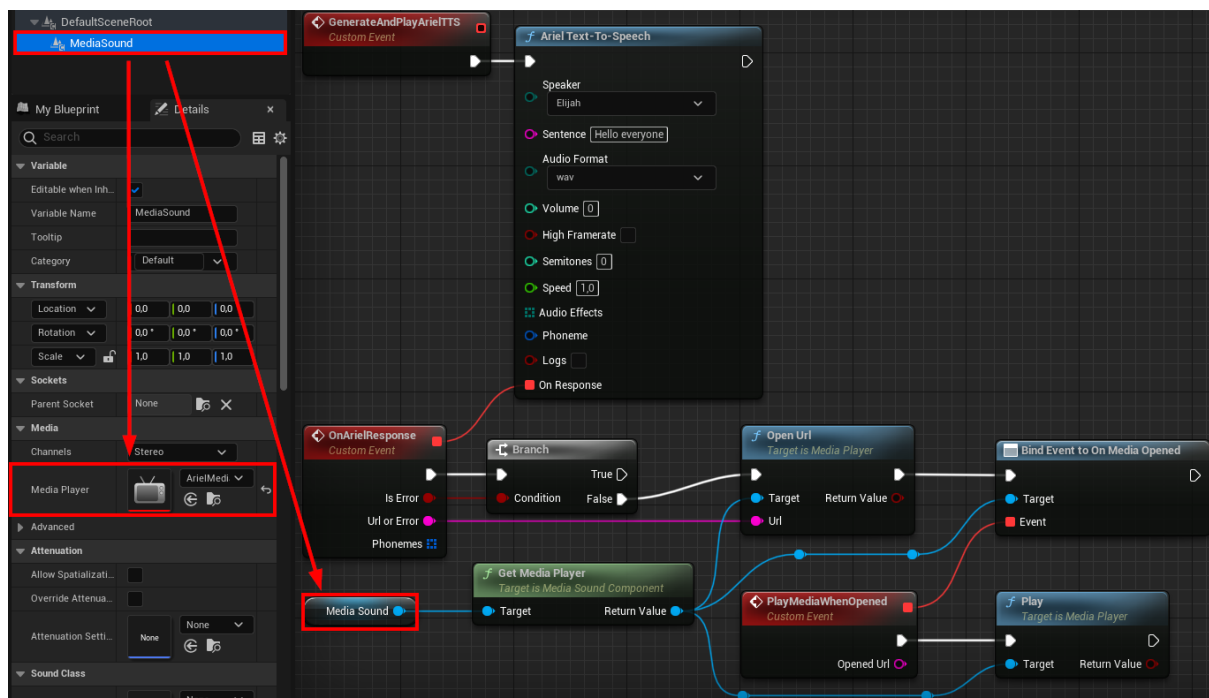
Is Error: Indicate if there was an error while processing the request.

URL or Error: The generated audio URL (if success) or the error message (if error).

Phonemes: *Optional.* The list of all phonemes for the sentence. You may break array elements in order to retrieve the phoneme, the time and the amplitude associated:



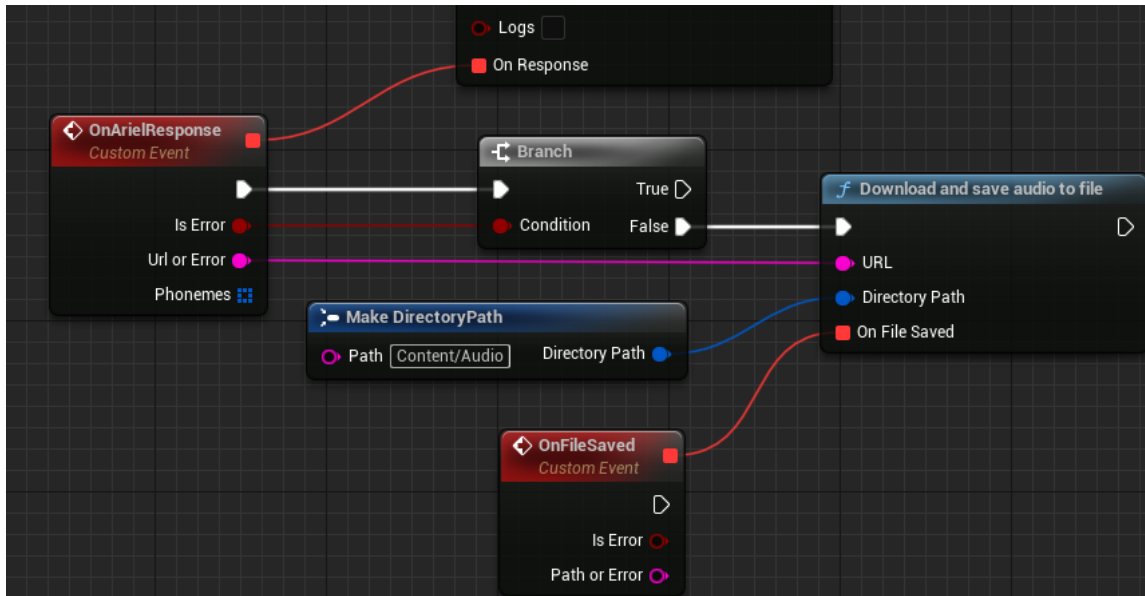
The generated audio URL can be passed to an Unreal “Media Player” or “Media steam” in order to be read at runtime. The image below show an example generating and playing the speech at runtime:



For more informations, please see: <https://blueprintue.com/blueprint/te9qcpy5/>

3.2. Download and save to file

This function takes the URL generated from the previous function (*Ariel Text-To-Speech*), downloads the audio and stores it in the specified directory. The response is sent through a delegate:



More information about Download function parameters:

URL: The URL of the audio file who needs to be downloaded.

Directory path: Define the directory where the downloaded file needs to be stored.

More information about Download return response:

You can bind and handle download response using nodes “Add custom event” or “Create event”.

Is Error: Indicate if there was an error while processing the request.

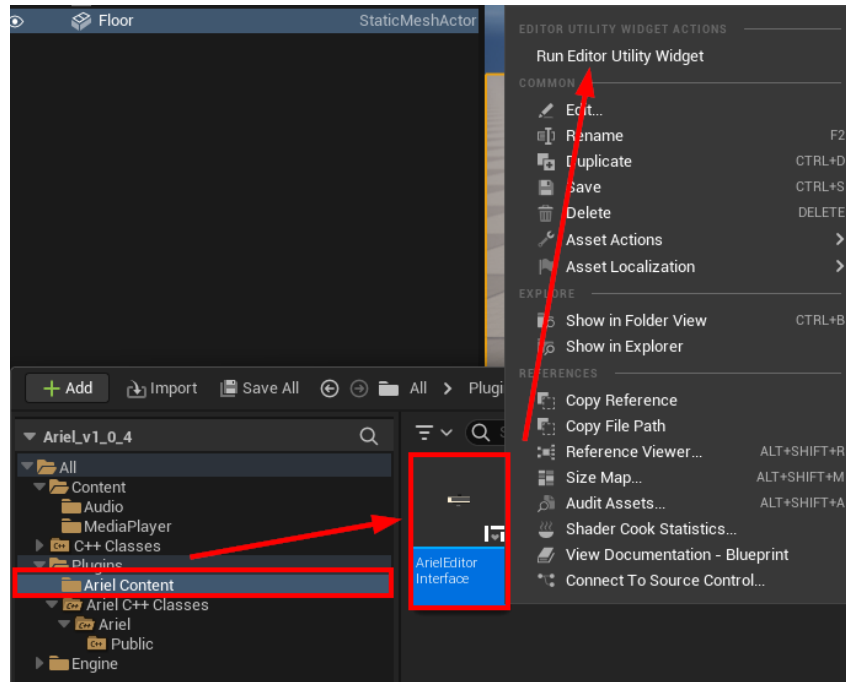
Path or Error: The audio file location (if successful, with filename) or the error message.

For more informations, please see: <https://blueprintue.com/blueprint/8hvffckk/>

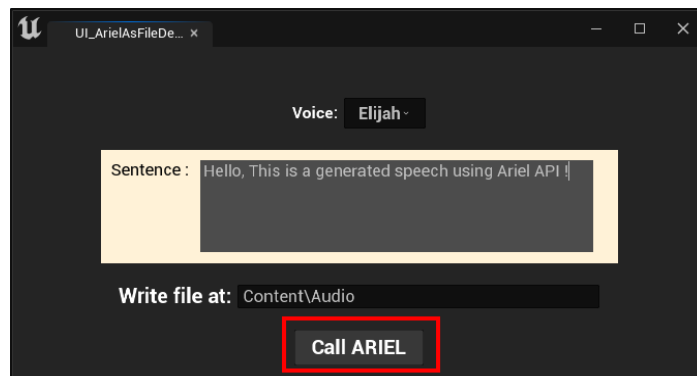
4. How to use the Ariel plug-in in Editor (with graphical interface)

If you want to generate audio sentences that will be saved as a file, you can use the “Editor Utility Widget” included in the plugin.

In the content browser, right click on “ArielEditorInterface” (located in *Plugins/Ariel Content/*) and choose “Run Editor Utility Widget” :

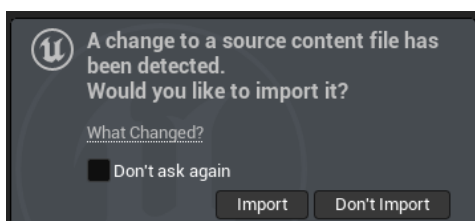


Fill all required informations (voice, sentence, directory, ...) and click “Call ARIEL”:



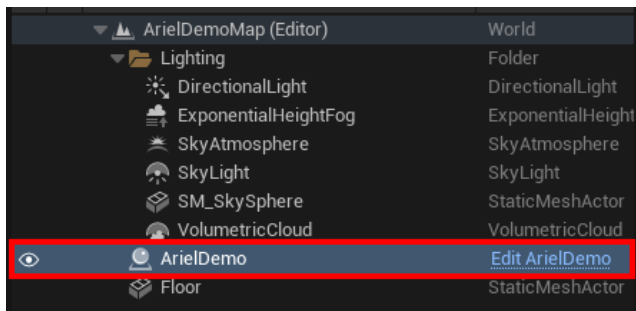
Wait for the API to generate audio and write it to a file. If you enter a path located in your unreal project, you should see a popup asking you if you want to import the file:

File written at: D:/XImmersion/Unreal Projects/PLUGIN_DEMO/Content/Audio/4d64ccfb-16ac-498f-aabb-ea28787b029d.wav

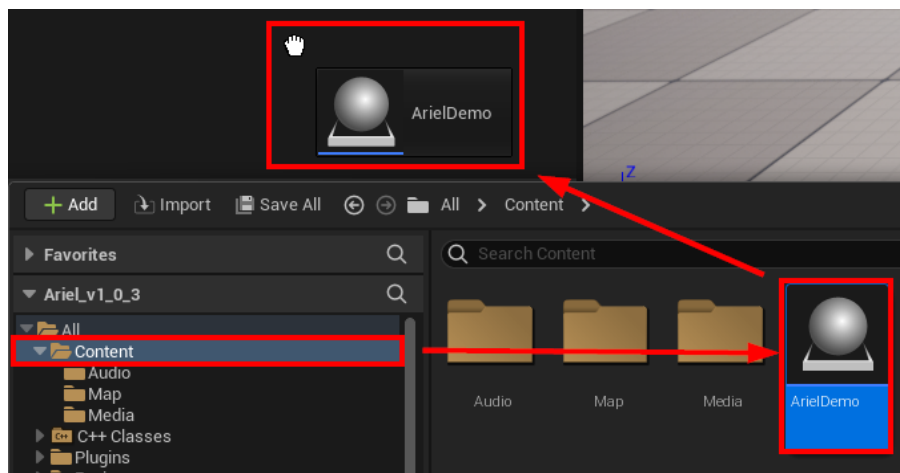


5. Demo (Runtime)

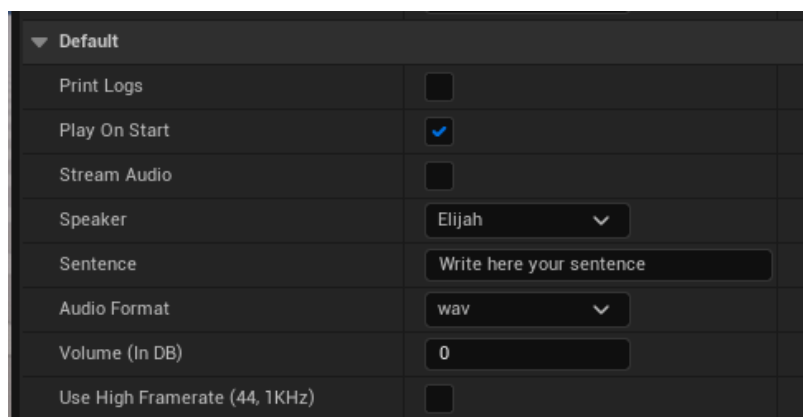
On the “ArielDemoMap” level, click on the Actor named “ArielDemo”:



You may also want to use the Actor in another level. To do it, simply drag and drop the Actor “ArielDemo” from the content browser (located in Content) in your level:



Once “ArielDemo” is selected in your level, you can define on the “details” pane all Ariel parameters (under *Default*):



You can find more information about every parameter in the chapter [Ariel Text-To-Speech](#).

In addition, there is two additional parameters:

Play on start: Indicate if the speech generation is made when the game starts.

Stream audio: Indicate if the audio should be played in streaming or downloaded.

6. Speaker list

This is the list of the Speakers currently supported by Ariel.

Premium voices :

Speaker name	Voice type	Language	Male/Female/Neutral
Oriane	Premium	English	Female
Dulhan	Premium	English	Male
Alioth	Premium	English	Male
Zenaya	Premium	English	Female
Ada	Premium	English	Female
Socrates	Premium	English	Male
David	Premium	English	Male
Tom	Premium	English	Male
Baldur	Premium	English	Male
Elizabeth	Premium	English	Female
Elen	Premium	English	Female
Riley	Premium	English	Neutral
Orc	Premium	English	Neutral
Urukhai	Premium	English	Neutral
Goblin	Premium	English	Neutral
HighElve	Premium	English	Female
DarkElve	Premium	English	Male
HalfElve	Premium	English	Female
WoodElve	Premium	English	Male
Dis	Premium	English	Female
Dwarf	Premium	English	Male

Nordman	Premium	English	Male
Duergar	Premium	English	Male

Standard voices :

Speaker name	Voice type	Language	Male/Female
Farah	Standard	Arabic	Female
Daiyu	Standard	Chinese (Mandarin)	Female
Emma	Standard	Danish	Female
Oscar	Standard	Danish	Male
Anke	Standard	Dutch	Female
Adriaan	Standard	Dutch	Male
Mia	Standard	English (Australian)	Female
Grace	Standard	English (Australian)	Female
Jack	Standard	English (Australian)	Male
Charlotte	Standard	English (British)	Female
Sophia	Standard	English (British)	Female
Elijah	Standard	English (British)	Male
Advika	Standard	English (Indian)	Female
Onkar	Standard	English (Indian)	Female
Elna	Standard	English (South African)	Female
Mary	Standard	English (US) - child	Female
Linda	Standard	English (US)	Female
Patricia	Standard	English (US)	Female
Barbara	Standard	English (US)	Female
Susan	Standard	English (US)	Female
Paul	Standard	English (US)	Male
Michael	Standard	English (US) - child	Male
William	Standard	English (US) - child	Male
Thomas	Standard	English (US)	Male

Speaker name	Voice type	Language	Male/Female
Aeron	Standard	English (Welsh)	Male
Capucine	Standard	French	Female
Alix	Standard	French	Female
Arnaud	Standard	French	Male
Stephanie	Standard	French (Canadian)	Female
Celine	Standard	French (Canadian)	Female
Maria	Standard	German	Female
Theresa	Standard	German	Female
Felix	Standard	German	Male
Anna	Standard	Icelandic	Female
Sigriour	Standard	Icelandic	Male
Gabriella	Standard	Italian	Female
Bella	Standard	Italian	Female
Lorenzo	Standard	Italian	Male
Rika	Standard	Japanese	Female
Tanaka	Standard	Japanese	Male
Ji-Ho	Standard	Korean	Female
Camilla	Standard	Norwegian	Female
Katarzyna	Standard	Polish	Female
Malgorzata	Standard	Polish	Female
Piotr	Standard	Polish	Male
Jan	Standard	Polish	Male
Tabata	Standard	Portuguese (Brazilian)	Female
Juliana	Standard	Portuguese (Brazilian)	Female
Pedro	Standard	Portuguese (Brazilian)	Male

Speaker name	Voice type	Language	Male/Female
Pati	Standard	Portuguese (European)	Female
Adriano	Standard	Portuguese (European)	Male
Alexandra	Standard	Romanian	Female
Inessa	Standard	Russian	Female
Viktor	Standard	Russian	Male
Francisca	Standard	Spanish (European)	Female
Margarita	Standard	Spanish (European)	Female
Mateo	Standard	Spanish (European)	Male
Leticia	Standard	Spanish (Mexican)	Female
Josefina	Standard	Spanish (US)	Female
Rosa	Standard	Spanish (US)	Female
Miguel	Standard	Spanish (US)	Male
Eva	Standard	Swedish	Female
Mesut	Standard	Turkish	Female
Angharad	Standard	Welsh	Female